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Propagation dynamics of malware in wireless sensor networks with node sleep-active scheduling based on simplicial complexes

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Abstract

Traditional studies of malware propagation in wireless sensor networks (WSNs) have predominantly focused on pairwise interactions. However, real-world contagion processes often involve higher-order interactions, in which multiple nodes jointly participate in infection events. To capture such group-based dynamics, this study proposes a malware propagation model that features higher-order interactions and node sleep-scheduling. Within the framework of the quenched mean-field (QMF) theory, we derive the node probability evolution equations and employ the mean-field (MF) approximation to obtain the two malware propagation thresholds related to bistability, together with the existence and stability conditions of the equilibrium points. Theoretical analysis reveals that the pairwise infection rate can induce the emergence of bistability. Furthermore, a Sobol-based sensitivity analysis is conducted to quantify the influence of each model parameter on the persistent propagation threshold, and the results indicate that the transition rates between sleep and active states together with the average degree of pairwise interactions play a dominant role in determining the persistent propagation threshold. The numerical simulations show that increasing the node activation rate, or decreasing the sleep rate, can induce a transition from a forward to a backward bifurcation, leading to bistability in which malware may either vanish or persist depending on the initial infection level.

1. Introduction

The Internet of Things (IoT), which has been extensively studied across various disciplines, has become an integral part of modern applications. As a fundamental component of the IoT perception layer, WSNs enable efficient data collection and serve as a bridge between the physical and digital domains [1, 2]. A wireless sensor network (WSN) consists of numerous low-power wireless sensors that collaboratively sense, collect, process, and transmit data over wireless links. This information is subsequently forwarded to one or more base stations for centralized analysis and processing [3, 4]. Consequently, WSNs have been widely deployed in diverse application domains, such as smart construction and industrial automation [5, 6], as well as in mission-critical scenarios including disaster monitoring and national defense [7].

The widespread application of WSNs has also increased their exposure to security threats, making malware propagation (e.g., malicious code such as viruses and worms) a prominent concern. Malicious code can infect sensor nodes through wireless communication channels such as Bluetooth and Wi-Fi, resulting in data corruption, privacy breaches, or node malfunction, and potentially causing substantial economic and operational losses [8, 9]. Therefore, establishing mathematical models to analyze the mechanisms of malware propagation in WSNs is of critical importance.

The study of network dynamics has significantly advanced epidemiological research on complex networks. Models derived from this framework are fundamental for investigating disease propagation, invasion dynamics, and the stability of epidemiological equilibria. The introduction of concepts such as small-

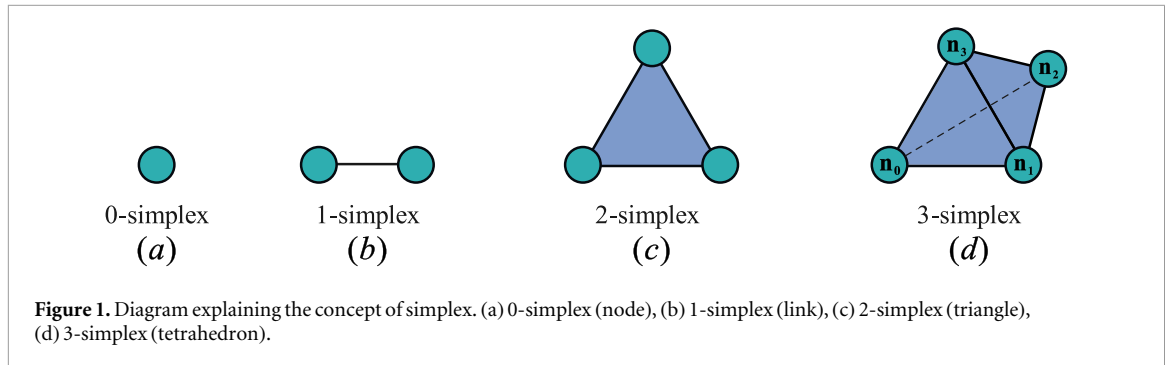
world networks [10] and scale-free networks [11] spurred the integration of epidemiology with complex network theory, leading to the adaptation of classical models such as Susceptible–Infectious–Susceptible (SIS) [12], Susceptible–Infected–Recovered (SIR) [13], and Susceptible–Exposed–Infected–Recovered (SEIR) [14] to these networked structures.

In traditional epidemiology on complex networks, diseases primarily spread to susceptible nodes through pairwise interactions between two nodes. However, real-world infections often occur in the context of group interactions, where multiple infectious sources simultaneously transmit to a susceptible node. For instance, in settings such as households, restaurants, or schools, interactions involving three or more individuals may jointly infect a susceptible individual. This interaction pattern is termed a ‘higher-order interaction’ [15]. From a network-structural perspective, the interaction patterns among multiple individuals give rise to what is referred to as a ‘higher-order network structure’ [16].

Current research primarily models such higher-order interactions using simplicial complexes. Iacopini *et al* first employed simplicial complexes to construct a social contagion model based on group interactions, in which contagion can occur through simplices of varying sizes. Their work revealed that higher-order interactions can induce discontinuous phase transitions and the emergence of bistable regions [17]. Cisneros-Velarde and Bullo analyzed a multi-group SIS epidemic model with higher-order interactions and rigorously established sufficient conditions for the emergence of bistability in heterogeneous group settings [18]. Kiss *et al* derived the exact Kolmogorov equations and their mean-field limits for SIS contagion dynamics on fully connected simplicial complexes, providing a rigorous analytical framework and revealing multistable phenomena induced by third-order interactions [19]. Yang *et al* constructed and analyzed an SIR model on a second-order simplicial complex, incorporating a nonlinear incidence rate as well as population birth and death dynamics, and identified the conditions under which the system exhibits bistability and backward bifurcations [20]. Bifurcation phenomena are also widely observed in other classes of nonlinear complex systems. For example, Lu *et al* [21] showed that time delays can induce Hopf bifurcations and qualitative changes in neural network dynamics, while subsequent studies demonstrated that fractional-order effects [22] and diffusion mechanisms [23] can further enrich bifurcation structures in large-scale networks.

The connection between malware propagation dynamics in WSNs and epidemiology on complex networks has attracted considerable scholarly attention. Indeed, malware spreading in WSNs exhibits strong similarities to disease transmission processes described by epidemiological models. Within this framework, wireless sensors are abstracted as nodes in a complex network, while malware is modeled as an infectious agent. Malware propagates by exploiting wireless communication channels used for data transmission during sensor operation, thereby infecting vulnerable nodes, whereas recovery mechanisms, such as antivirus disinfection or hardware resets, can restore infected sensors to their normal operational state [24]. Consequently, the analytical tools developed in epidemiology on complex networks are well suited for analyzing malware propagation in WSNs. In most existing studies, malware spreading is modeled through peer-to-peer infection mechanisms, whereby a susceptible node becomes infected via one-to-one contact with infected neighbors. A representative early contribution by Tang and Mark [25] introduced epidemic modeling into the analysis of virus spreading in wireless sensor networks, thereby establishing a foundational pairwise interaction framework. Building on this paradigm, Shen *et al* [26] further developed a compartmental epidemic model to characterize malware diffusion dynamics in heterogeneous WSNs, demonstrating that pairwise infection assumptions can be extended to incorporate more realistic network heterogeneity and node-level characteristics.

Due to their energy consumption and limited battery power, wireless sensors are generally designed with periodic sleep-scheduling mechanisms. In such schemes, nodes alternate between active and sleeping states, where sleeping nodes suspend communication and sensing activities to reduce energy expenditure. This design philosophy was first systematically introduced through energy-efficient Medium Access Control (MAC) protocols for WSNs [27] and was later extended to low-duty-cycled sensor networks with coordinated and random sleep strategies [28]. As a result, sleep scheduling has become a fundamental characteristic of wireless sensor networks and plays a crucial role in both network performance and security modeling. For instance, Upadhyay and Kumari proposed a Susceptible–Infected–Terminally Infected–Recovered (SITR) model with a terminal (T) state to represent sensor dormancy. Using a **cubic-type** functional response, their bifurcation analysis identified key factors that can induce network instability [29]. To account for spatial effects, Shakya *et al* developed a Susceptible–Exposed–Infected–Recovered–Susceptible (SEIRS) model for event-driven WSNs, demonstrating that the spreading threshold depends on the spatial correlation between nodes [30]. Liu *et al* proposed a novel Susceptible–Infected–Low-energy–Susceptible (SILS) epidemic model for wireless rechargeable sensor networks. Through stability analysis combined with differential game theory, they not only revealed a positive correlation between the charging rate and the basic reproduction number, but also derived optimal attack–defense strategies to suppress malware propagation and reduce network costs [31].



However, the core functions of WSNs (such as environmental sensing or target tracking) rely on collaborative operations among multiple nodes. For example, a cluster head may need to fuse data collected from several neighboring sensors to support decision making. Malware can exploit such collaboration mechanisms: when a group of active-infected neighboring nodes participates in a joint data transmission, they may collectively infect an active-susceptible sensor. In this case, the infection process depends on the collective state or behavior of a group of nodes rather than on isolated pairwise contacts. This observation implies that, analogous to epidemic spreading with higher-order interactions, malware propagation in WSNs can also exhibit higher-order characteristics.

Although SIS-type models incorporating sleep-scheduling mechanisms for malware propagation in WSNs have been investigated in previous studies [32], SIS models that simultaneously account for both sleep-scheduling and higher-order interactions remain insufficiently explored. Motivated by this gap, we construct an SIS model that integrates probabilistic node sleep-scheduling with higher-order interaction mechanisms. In this paper, we make the following main contributions:

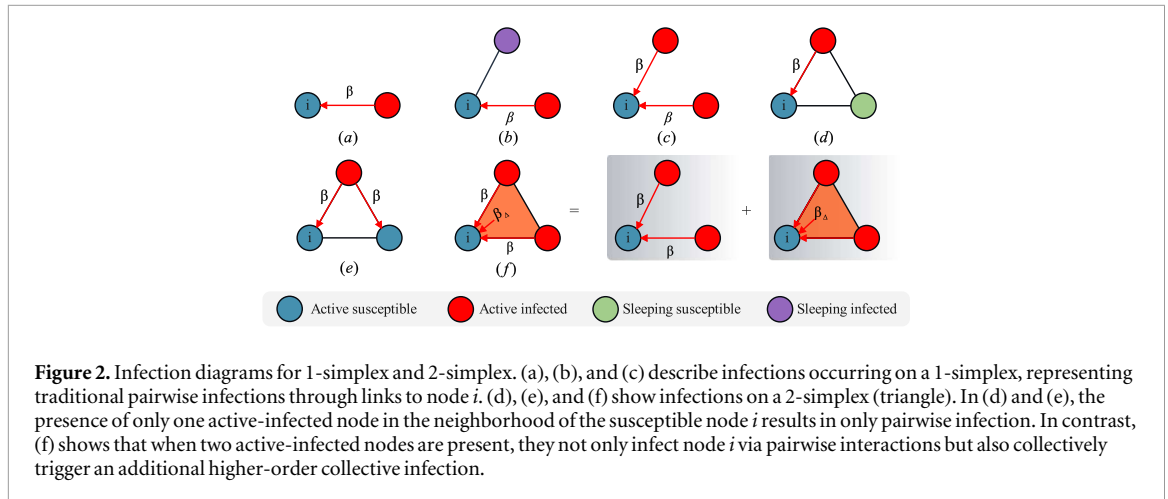
- (1) This paper employs the QMF method to establish a malware propagation model for WSNs incorporating node sleep scheduling and higher-order interactions. The propagation dynamics are further simulated using the Monte Carlo (MC) method, and the two sets of results are systematically compared. The numerical results indicate that, as the pairwise infection rate increases, the agreement between the QMF-based analysis and MC simulations improves progressively, which supports the accuracy and rationality of the proposed model.
- (2) This study adopts the MF approximation to reduce the dimensionality of the model, and systematically analyzes the existence and stability of equilibrium points. Based on this analysis, the corresponding propagation thresholds are derived. The theoretical results demonstrate that, under certain conditions, variations in the pairwise infection rate can lead to qualitatively different steady-state outcomes.
- (3) Numerical simulations reveal that increasing the active conversion rate or decreasing the sleep conversion rate can induce a backward bifurcation in the system. These parameters exert distinct influences on the sleeping and active infection densities. Moreover, the results show that the initial infection densities play a crucial role in determining the system's convergence to different steady states.

The organization of the content in this paper proceeds as follows. Section 2 primarily elucidates the concept of simplicial complexes, the meaning of sleep-active scheduling, and the propagation behavior of the model. Section 3 constructs the model using the QMF method and verifies the accuracy and rationality of the model through MC simulations. Section 4 primarily analyzes the basic reproduction number, the existence, stability, and corresponding conditions of equilibrium points, and derives the two thresholds related to bistability. Section 5 performs a sensitivity analysis of the model to quantify the influence of each parameter on the persistence threshold. Section 6 corroborates the theoretical findings with a series of numerical simulations and analyzes the impact of individual parameters on the backward bifurcation. Section 7 encapsulates the entire paper and delineates prospective avenues for further research.

2. Model description

2.1. The description of simplicial complex

A simplicial complex is a mathematical structure composed of one or more simplices, used to represent higher-order interactions among nodes. A set of $d + 1$ fully interacting nodes, $\{n_0, n_1, \dots, n_d\}$, forms a d -simplex, denoted by $\alpha_d = [n_0, n_1, \dots, n_d]$, where d is the dimension of the simplex [33]. As illustrated in figure 1, typical examples include a 0-simplex (node), a 1-simplex (link), a 2-simplex (triangle), and a 3-simplex (tetrahedron). The dimension of a simplicial complex is defined as the maximum dimension of its constituent simplices. Any



sub-simplex formed by a subset of $h + 1$ nodes of a d -simplex (where $h \leq d$) is referred to as an h -face of that simplex. Crucially, a simplicial complex must be closed under the inclusion of faces; that is, if a simplex belongs to the complex, all its faces must also be included.

For example, the 3-simplex $\alpha_3 = [n_0, n_1, n_2, n_3]$ shown in figure 1(d) contains four 0-faces, six 1-faces, and four 2-faces. A fundamental property of a simplicial complex is that it is closed under face inclusion: if a simplex belongs to the complex, then all of its faces must also belong to the complex.

From the structural perspective of WSNs, the average degree $\langle k \rangle$ characterizes the number of pairwise communication links per node, while the generalized average degree $\langle k_\Delta \rangle$ measures the abundance of higher-order interaction groups (2-simplices) in which a node participates. The feasible range of $\langle k \rangle$ is determined by the trade-off between connectivity and wireless interference: connectivity theory imposes a lower bound to ensure the existence of a giant connected component [34], whereas excessively large degrees may lead to severe packet collisions and throughput degradation [35]. In contrast, $\langle k_\Delta \rangle$ serves as a structural parameter that controls the strength of higher-order interactions induced by broadcast, cooperative, or group-based communication patterns. While in many practical WSN deployments higher-order interactions typically coexist with pairwise connectivity, their relative prevalence may vary significantly depending on network protocols, traffic patterns, and application scenarios.

2.2. Sleep-active scheduling mechanism

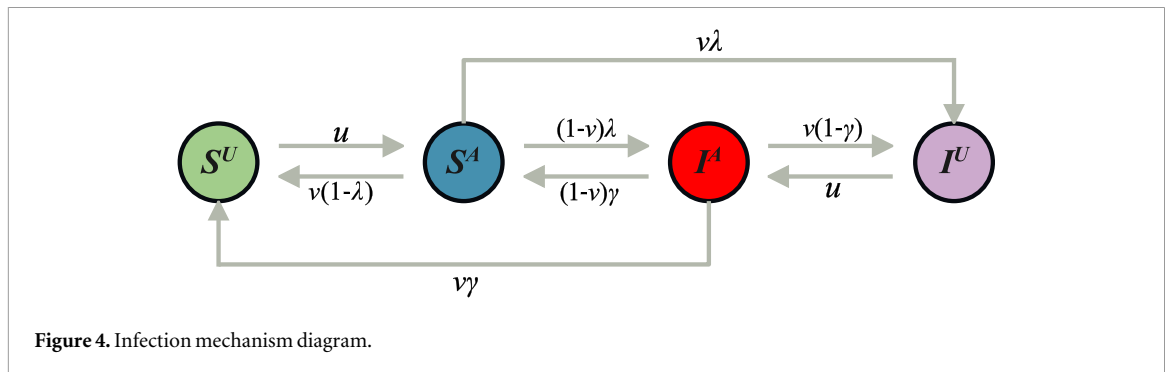
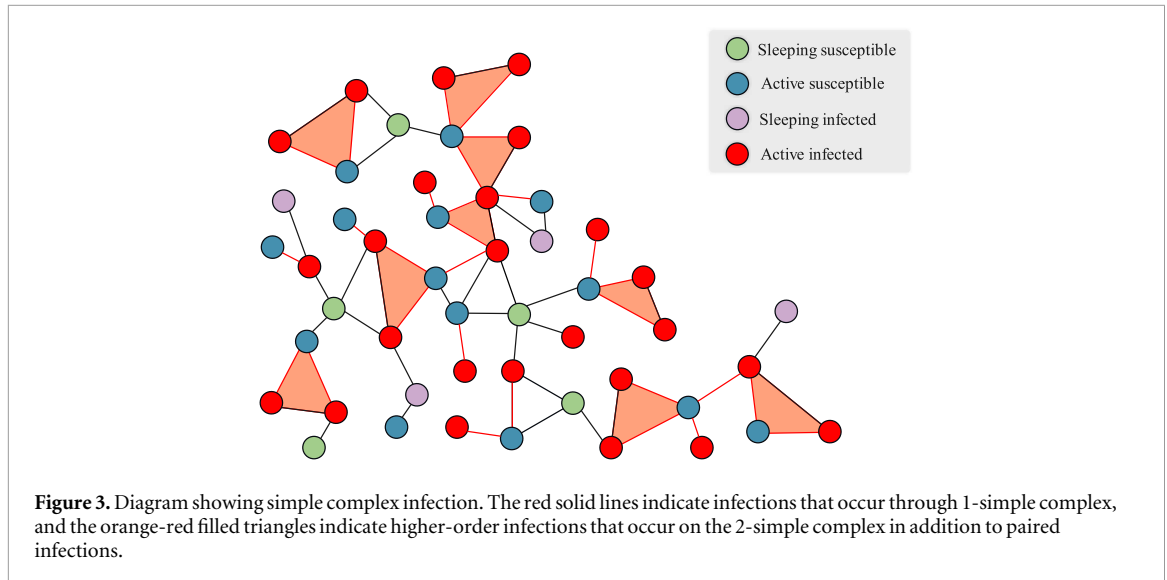
WSNs are assumed to operate under a sleep-active scheduling mechanism. Each sensor node can be in one of two states: sleeping, denoted by U , or active, denoted by V . A node switches from the sleep state U to the active state V with transition rate u , and from the active state V back to the sleep state U with transition rate v .

The sleep-active transition rates u and v are directly determined by duty-cycling mechanisms implemented at the MAC layer, such as those specified in the IEEE 802.15.4 standard. Specifically, u governs the wake-up frequency of sleeping nodes, whereas v characterizes the rate at which active nodes transition into the sleep state. In energy-constrained WSNs, the effective duty cycle (DC), defined as $\frac{u}{u+v}$, is typically maintained at a low level to prolong network lifetime [36].

2.3. The description of spread

We adopt an SIS epidemiological spreading mechanism to model malware propagation in WSNs. Each sensor node is assumed to be in one of two epidemiological states: susceptible (S) or infected (I). Incorporating the sleep-active scheduling mechanism, the state space is extended to four possible states, namely sleeping-susceptible (S^U), active-susceptible (S^A), sleeping-infected (I^U), and active-infected (I^A). Only nodes in the active state are involved in infection and recovery processes. Nodes in sleeping states are excluded from both infection and recovery and can only transition to the active state. Active-infected nodes recover with rate γ , where the recovery rate γ reflects the effectiveness of malware mitigation mechanisms in real-world WSNs, such as antivirus scanning, firmware reinstallation, or manual system resets. Consequently, active nodes are subject to infection, recovery, and sleep-active transitions.

In the simplicial-complex-based higher-order network, higher-order infection is defined by a specific rule: for a given simplex of any order, a higher-order infection event is triggered on an active-susceptible node if, and only if, all other nodes within that simplex are in the active-infected state. This higher-order infection acts in addition to any pairwise infections. Taking figure 2 as an example, panels (a) and (b) illustrate pairwise



infection on a 1-simplex (link), where an active-infected node contributes a pairwise infection rate β to node i ; sleeping-infected nodes do not participate in malware propagation. Panels figures 2(d) and (e) depict infection processes on a 2-simplex that occur exclusively through pairwise interactions, and sleeping-susceptible nodes cannot be infected. More notably, figure 2(f) illustrates higher-order infection on a 2-simplex (triangle). When the other two nodes in the simplex containing the active-susceptible node i are both in the active-infected state, they collectively induce an additional triangular higher-order infection on node i with rate β_{Δ} , on top of the pairwise infections transmitted through the links.

By combining pairwise and higher-order infection mechanisms, the propagation dynamics of malware can be described more accurately. Figure 3 illustrates a second-order simplicial complex undergoing internal infection, where infections transmitted through traditional links (pairwise interactions) are represented by red lines, while the additional triangular infections arising from higher-order interactions are indicated by the orange-shaded regions.

The pairwise infection rate β reflects the effectiveness of point-to-point (unicast) malware transmission and is closely related to physical-layer link quality indicators, such as the packet error rate (PER), signal-to-noise ratio (SNR), and transmission power. In contrast, the higher-order infection rate β_{Δ} captures the vulnerability associated with group-based communication patterns enabled by the broadcast nature of wireless channels. Due to the wireless broadcast advantage, a single transmission from an infected node may be simultaneously received by multiple neighboring nodes forming a simplicial clique, thereby inducing higher-order contagion effects.

Let λ denote the total infection rate exerted on a susceptible node by all simplices containing it. The corresponding malware propagation mechanism is illustrated in figure 4.

For clarity and ease of reference, all model parameters and their corresponding descriptions are summarized in table 1.

Remark 1. In practical applications of the propagation model, the admissible ranges of parameters in system (5) can be restricted or determined based on their mathematical definitions and modeling assumptions. The parameters u and v are continuous-time transition rates between sleep and active states and are constrained by

Table 1. List of notations, parameters, and variables.

Notations, parameters, and variables	Definitions
N	The total number of nodes in the simplicial complex
u	Active conversion rate, i.e., the rate at which a node switches from sleep to active state
v	Sleep conversion rate, i.e., the rate at which a node switches from active to sleep state
γ	Recovery rate
β	The infection rate from pairwise (1-simplex) interactions
β_{Δ}	The infection rate from triplewise (2-simplex) interactions
β_q	The infection rate from q -th order interactions
η	Rescaled pairwise infection rate
λ	The total infection rate of the active susceptible node
$\langle k \rangle$	The average degree, i.e., the average number of 1-simplices (edges) incident to a node
$\langle k_{\Delta} \rangle$	The average generalized degree of order 2, i.e., the average number of 2-simplices (triangles) incident to a node
$\langle k_q \rangle$	The average generalized degree of order q , i.e., the average number of q -simplices incident to a node
Λ_i	The set of 1-simplices (links) to which node i belongs
Ω_i	The set of 2-simplices (triangles) to which node i belongs
Γ_i^q	The set of q -simplices to which node i belongs

the characteristic time scales of duty-cycling mechanisms, which can be quantified from prescribed scheduling periods or average state sojourn times. The infection rates β and β_{Δ} , together with the recovery rate γ , are nonnegative rate parameters governing transmission and removal processes, whose effective magnitudes can be inferred from the relative frequencies of infection and recovery events. The structural quantities $\langle k \rangle$ and $\langle k_{\Delta} \rangle$ are determined by the underlying simplicial complex construction and network connectivity and can be obtained from network topology statistics.

3. Model formulation

The QMF method is widely used to study propagation dynamics in a variety of real-world networks. Here, we employ the QMF method to establish an SIS model incorporating higher-order infection interactions and a sleep-scheduling mechanism, based on the malware propagation mechanism in WSNs illustrated in figure 4. The probabilities that node i is in the sleeping susceptible state, active susceptible state, sleeping infected state, and active infected state at time t are denoted by $S_i^U(t)$, $S_i^A(t)$, $I_i^U(t)$, and $I_i^A(t)$, respectively. These quantities collectively represent the state densities of node i in the network. Accordingly, we formulate the QMF model and further simplify it to obtain equation (1):

$$\begin{cases} \frac{dS_i^U(t)}{dt} = -uS_i^U(t) + v(1 - \lambda_i)S_i^A(t) + v\gamma I_i^A(t) \\ \frac{dS_i^A(t)}{dt} = uS_i^U(t) + (1 - v)\gamma I_i^A(t) - v(1 - \lambda_i)S_i^A(t) - (1 - v)\lambda_i S_i^A(t) - v\lambda_i S_i^A(t) \\ \frac{dI_i^U(t)}{dt} = -uI_i^U(t) + v\lambda_i S_i^A(t) + v(1 - \gamma)I_i^A(t) \\ \frac{dI_i^A(t)}{dt} = -(1 - v)\gamma I_i^A(t) - v(1 - \gamma)I_i^A(t) - v\gamma I_i^A(t) + uI_i^U(t) + (1 - v)\lambda_i S_i^A(t), \end{cases} \quad (1)$$

where

$$\lambda_i = \sum_q \beta_q \sum_{\{j_1, \dots, j_q\} \in \Gamma_i^q} I_{j_1}^A(t) I_{j_2}^A(t) \cdots I_{j_q}^A(t)$$

represents the total infection rate exerted on an active susceptible node S_i^A by all simplices containing it.

Substituting the above expression for λ_i into equation (1) and simplifying yields the system of equations given in equation (2):

$$\left\{ \begin{aligned} \frac{dS_i^U(t)}{dt} &= -uS_i^U(t) + vS_i^A(t) + v\gamma I_i^A(t) \\ &\quad - v \sum_q \beta_q S_i^A(t) \sum_{\{j_1, \dots, j_q\} \in \Gamma_i^q} I_{j_1}^A(t) I_{j_2}^A(t) \cdots I_{j_q}^A(t) \\ \frac{dS_i^A(t)}{dt} &= uS_i^U(t) - vS_i^A(t) + (1-v)\gamma I_i^A(t) \\ &\quad - (1-v) \sum_q \beta_q S_i^A(t) \sum_{\{j_1, \dots, j_q\} \in \Gamma_i^q} I_{j_1}^A(t) I_{j_2}^A(t) \cdots I_{j_q}^A(t) \\ \frac{dI_i^U(t)}{dt} &= -uI_i^U(t) + v(1-\gamma)S_i^A(t) \\ &\quad + v \sum_q \beta_q S_i^A(t) \sum_{\{j_1, \dots, j_q\} \in \Gamma_i^q} I_{j_1}^A(t) I_{j_2}^A(t) \cdots I_{j_q}^A(t) \\ \frac{dI_i^A(t)}{dt} &= uI_i^U(t) + (v\gamma - v - \gamma)I_i^A(t) \\ &\quad + (1-v) \sum_q \beta_q S_i^A(t) \sum_{\{j_1, \dots, j_q\} \in \Gamma_i^q} I_{j_1}^A(t) I_{j_2}^A(t) \cdots I_{j_q}^A(t). \end{aligned} \right. \quad (2)$$

The strict requirement for higher-order infection is that all nodes within a simplex, except for a single susceptible node, must be infected. As a result, the probability of such an event decreases rapidly as the simplex order increases. Consequently, the contribution of high-order simplices to the overall infection dynamics is often negligible.

Based on this observation, we focus our subsequent analysis on a system defined on a 2-simplicial complex, considering only pairwise and 2-simplex (triangular) interactions. In this case, equation (2) reduces to the form given in equation (3):

$$\left\{ \begin{aligned} \frac{dS_i^U(t)}{dt} &= -uS_i^U(t) + vS_i^A(t) + v\gamma I_i^A(t) \\ &\quad - v\beta_S^A(t) \sum_{j \in \Lambda_i} I_j^A(t) - v\beta_\Delta^A(t) \sum_{\{j_1, j_2\} \in \Omega_i} I_{j_1}^A(t) I_{j_2}^A(t) \\ \frac{dS_i^A(t)}{dt} &= uS_i^U(t) - vS_i^A(t) + (1-v)\gamma I_i^A(t) \\ &\quad - (1-v)\beta_S^A(t) \sum_{j \in \Lambda_i} I_j^A(t) - (1-v)\beta_\Delta^A(t) \sum_{\{j_1, j_2\} \in \Omega_i} I_{j_1}^A(t) I_{j_2}^A(t) \\ \frac{dI_i^U(t)}{dt} &= -uI_i^U(t) + v(1-\gamma)S_i^A(t) \\ &\quad + v\beta_S^A(t) \sum_{j \in \Lambda_i} I_j^A(t) + v\beta_\Delta^A(t) \sum_{\{j_1, j_2\} \in \Omega_i} I_{j_1}^A(t) I_{j_2}^A(t) \\ \frac{dI_i^A(t)}{dt} &= uI_i^U(t) + (v\gamma - v - \gamma)I_i^A(t) \\ &\quad + (1-v)\beta_S^A(t) \sum_{j \in \Lambda_i} I_j^A(t) + (1-v)\beta_\Delta^A(t) \sum_{\{j_1, j_2\} \in \Omega_i} I_{j_1}^A(t) I_{j_2}^A(t). \end{aligned} \right. \quad (3)$$

The Random Simplicial Complex (RSC) framework provides a natural and systematic approach for constructing networked systems composed of simplices of different dimensions, thereby enabling the explicit modeling of higher-order interactions among multiple nodes. In this work, we employ the RSC method proposed by Iacopini *et al* to generate simplicial complexes with prescribed statistical properties [17]. Specifically, we consider a system consisting of $N = 2000$ nodes, an average degree of $\langle k \rangle = 20$, and a generalized average degree of $\langle k_\Delta \rangle = 6$. It is crucial to note that the formation of a 2-simplex intrinsically creates three 1-simplices on its boundary. Therefore, to achieve the desired total average degree $\langle k \rangle$, the probability p_1 of generating independent 1-simplices between pairs of nodes must be adjusted to account for the 1-simplices induced by higher-order structures.

As shown in Algorithm 1, given the network size N and the target degrees, the generation process involves two probabilities: p_1 , which controls the creation of 1-simplices, and p_2 , which represents the probability of forming a 2-simplex among any triplet of nodes. Following the RSC method proposed by Iacopini *et al* [17], these probabilities are defined as

$$p_1 = \frac{\langle k \rangle - 2\langle k_\Delta \rangle}{(N-1) - 2\langle k_\Delta \rangle}, \quad p_2 = \frac{2\langle k_\Delta \rangle}{(N-1)(N-2)}.$$

The detailed generative procedure is summarized in algorithm 1. The algorithm proceeds in two steps: first, random 1-simplices are generated with probability p_1 ; second, random 2-simplices are generated with

probability p_2 , while the simplicial closure property is automatically enforced by adding any missing boundary edges.

On this simplicial complex, we validate the proposed model by comparing the results obtained from the constructed QMF model equation (3) on the RSC with those obtained from Monte Carlo (MC) simulations. To facilitate numerical analysis, a rescaled pairwise infection rate is introduced as $\eta = \langle k \rangle \beta / \gamma$. The MC results are averaged over multiple independent realizations. As shown in figure 5, the agreement between the QMF predictions and MC simulations improves with increasing rescaled pairwise infection rate, thereby validating the accuracy of the proposed model.

Algorithm 1. Generation of random simplicial complex with prescribed average degrees

Input: Number of nodes N ; the average degree $\langle k \rangle$; the average generalized degree $\langle k_\Delta \rangle$

Output: A random simplicial complex $\mathcal{K}$

1 Compute 1-simplex generation probability:

$$p_1 = \frac{\langle k \rangle - 2\langle k_\Delta \rangle}{(N-1) - 2\langle k_\Delta \rangle}$$

Compute 2-simplex generation probability:

$$p_2 = \frac{2\langle k_\Delta \rangle}{(N-1)(N-2)}$$

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2 Initialize node set  $\mathcal{V} = \{1, 2, \dots, N\}$ ;
3 Initialize empty 1-simplices set  $\Lambda$  (edge set) and 2-simplices set  $\Omega$ ;
4 Step 1: Generate 1-simplices (edges);
5 for  $i = 1$  to  $N$  do
6   for  $j = i + 1$  to  $N$  do
7     Generate a random number  $r \sim U(0, 1)$ ;
8     if  $r < p_1$  then
9       Add edge  $(i, j)$  to  $\Lambda$ ;
10 Step 2: Generate 2-simplices;
11 for  $i = 1$  to  $N$  do
12   for  $j = i + 1$  to  $N$  do
13     for  $k = j + 1$  to  $N$  do
14       Generate a random number  $r \sim U(0, 1)$  with high precision;
15       if  $r < p_2$  then
16         Add 2-simplex  $(i, j, k)$  to  $\Omega$ ;
17         // Ensure closure property of simplicial complex
18         if  $(i, j) \notin \Lambda$  then
19           Add edge  $(i, j)$  to  $\Lambda$ ;
20         if  $(i, k) \notin \Lambda$  then
21           Add edge  $(i, k)$  to  $\Lambda$ ;
22         if  $(j, k) \notin \Lambda$  then
23           Add edge  $(j, k)$  to  $\Lambda$ ;
23 return  $\mathcal{K} = (\mathcal{V}, \Lambda, \Omega)$ 

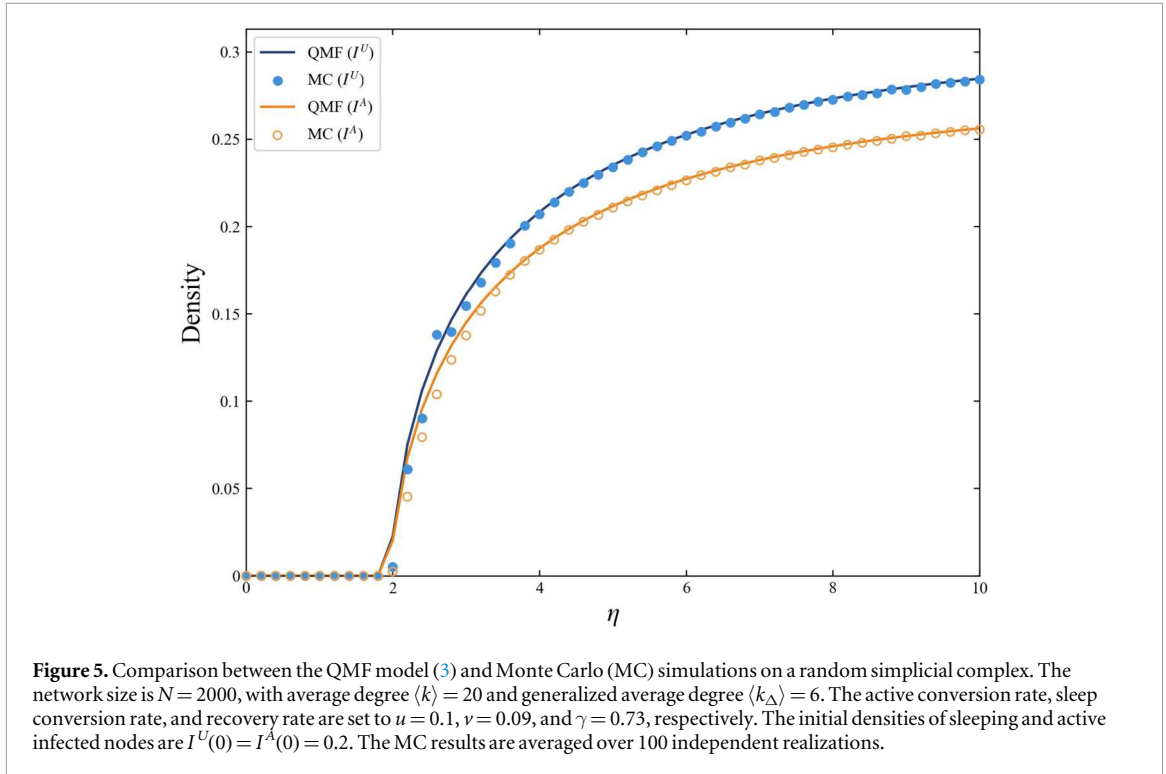
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4. Dynamic analysis of the model

The QMF model, described by system (3), is formulated to characterize malware propagation with higher-order interactions and a sleep-scheduling mechanism. However, system (3) is structurally complex, which makes its dynamical properties difficult to analyze directly.

In this work, we focus on macroscopic dynamical properties, including thresholds, equilibria, stability, and bifurcation behavior, rather than on node-level fluctuations. From a practical network perspective, wireless sensor networks are typically deployed with relatively high node density and redundant communication links to ensure sensing coverage and network connectivity. In such configurations, local structural fluctuations and node-specific correlations tend to be averaged out at the population level, making an average-degree-based description suitable for capturing macroscopic system behavior. Under the homogeneous mixing assumption, local pairwise and higher-order dynamical correlations are approximated by their population averages, which allows the QMF formulation to be further reduced to a MF approximation.

Although the MF approximation neglects microscopic correlations retained in the QMF description, it preserves the macroscopic effects of network topology and higher-order interactions through the average degree $\langle k \rangle$ and the average generalized degree $\langle k_\Delta \rangle$. Similar MF reductions have been widely adopted in recent



studies of epidemic spreading on simplicial complexes and have been shown to capture key qualitative features such as bistability and backward bifurcation [37, 38]. Motivated by these works, we adopt the MF approximation to obtain a tractable low-dimensional system, given by equation (4):

$$\begin{cases} \frac{dS^U(t)}{dt} = -uS^U(t) + vS^A(t) + v\gamma I^A(t) \\ \quad - v\beta \langle k \rangle S^A(t)I^A(t) - v\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2 \\ \frac{dS^A(t)}{dt} = uS^U(t) - vS^A(t) + (1-v)\gamma I^A(t) \\ \quad - (1-v)\beta \langle k \rangle S^A(t)I^A(t) - (1-v)\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2 \\ \frac{dI^U(t)}{dt} = -uI^U(t) + v(1-\gamma)I^A(t) \\ \quad + v\beta \langle k \rangle S^A(t)I^A(t) + v\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2 \\ \frac{dI^A(t)}{dt} = uI^U(t) + (v\gamma - v - \gamma)I^A(t) \\ \quad + (1-v)\beta \langle k \rangle S^A(t)I^A(t) + (1-v)\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2. \end{cases} \quad (4)$$

Given that $S^U(t) + S^A(t) + I^U(t) + I^A(t) = 1$, the dimensionality of system equation (4) can be further reduced, yielding the simplified system shown in equation (5):

$$\begin{cases} \frac{dS^A(t)}{dt} = u - (u+v)S^A(t) - uI^U(t) + (\gamma - u - v\gamma)I^A(t) \\ \quad - (1-v)\beta \langle k \rangle S^A(t)I^A(t) - (1-v)\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2 \\ \frac{dI^U(t)}{dt} = -uI^U(t) + v(1-\gamma)I^A(t) \\ \quad + v\beta \langle k \rangle S^A(t)I^A(t) + v\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2 \\ \frac{dI^A(t)}{dt} = uI^U(t) + (v\gamma - v - \gamma)I^A(t) \\ \quad + (1-v)\beta \langle k \rangle S^A(t)I^A(t) + v\beta_{\Delta} \langle k_{\Delta} \rangle S^A(t)(I^A(t))^2. \end{cases} \quad (5)$$

4.1. Disease-free equilibrium and stability analysis

The disease-free equilibrium represents a steady state of the WSN in which malware is completely absent from the network. This equilibrium, denoted by E_0 , is obtained by setting the time derivatives in system (5) to zero, namely, $\frac{dS^A(t)}{dt} = 0$, $\frac{dI^U(t)}{dt} = 0$, and $\frac{dI^A(t)}{dt} = 0$, which leads to the algebraic equations given in equation (6):

$$\begin{cases} u - (u + v)S^A(t) - uI^U(t) + (\gamma - u - v\gamma)I^A(t) - (1 - v)\beta\langle k \rangle S^A(t)I^A(t) \\ \quad - (1 - v)\beta_\Delta\langle k_\Delta \rangle S^A(t)(I^A(t))^2 = 0 \\ -uI^U(t) + v(1 - \gamma)I^A(t) + v\beta\langle k \rangle S^A(t)I^A(t) + v\beta_\Delta\langle k_\Delta \rangle S^A(t)(I^A(t))^2 = 0 \\ uI^U(t) + (v\gamma - v - \gamma)I^A(t) + (1 - v)\beta\langle k \rangle S^A(t)I^A(t) + (1 - v)\beta_\Delta\langle k_\Delta \rangle S^A(t)(I^A(t))^2 = 0. \end{cases} \quad (6)$$

By substituting $I^U(t) = I^A(t) = 0$ into system (6), we obtain the disease-free equilibrium $E_0 = \left(\frac{u}{u+v}, 0, 0\right)$ of system (5).

The basic reproduction number, denoted by R_0 , is a fundamental threshold that quantifies the transmission potential of malware in WSNs. It is defined as the expected number of secondary infections generated by a single infected node introduced into a completely susceptible population. The value of R_0 serves as a stability criterion for the disease-free equilibrium E_0 : E_0 is locally asymptotically stable if $R_0 < 1$ and unstable if $R_0 > 1$ [39].

Theorem 4.1. *The stability of the disease-free equilibrium E_0 is determined by the invasion threshold*

$$\beta_{c1} = \frac{\gamma(u + v)}{u\langle k \rangle}.$$

Specifically, E_0 is locally asymptotically stable for $\beta < \beta_{c1}$ and becomes unstable for $\beta > \beta_{c1}$.

Proof. The detailed proof is provided in appendix A.1. □

Remark 2. The disease-free equilibrium E_0 corresponds to a normal operating state of the wireless sensor network in which malware is completely eliminated. The invasion threshold β_{c1} therefore defines a security margin: maintaining $\beta < \beta_{c1}$ guarantees that malware cannot successfully invade the network. From a control perspective, increasing the sleep conversion rate v or decreasing the activation rate u provides an effective strategy for preserving this secure operating state.

Remark 3. It is worth noting that the derived invasion threshold β_{c1} , which is associated with the basic reproduction number R_0 , depends solely on the pairwise infection rate β and the sleep–active transition parameters u and v , while the higher-order infection rate β_Δ does not appear explicitly. From the perspective of linear stability analysis, the invasion threshold characterizes the local stability of the disease-free equilibrium, where the infection density is infinitesimal. Since higher-order infection mechanisms require the simultaneous presence of multiple infected neighbors (e.g., forming a 2-simplex), the corresponding infection force scales nonlinearly with the infection density. Consequently, during the linearization around the disease-free equilibrium, these higher-order nonlinear terms vanish, leaving only the linear pairwise infection terms to determine the initial growth condition. From the standpoint of the underlying propagation dynamics, this implies that during the initial invasion phase, infected nodes are sparse and the probability of forming infected cliques is negligible compared to that of pairwise contacts. As a result, the early-stage invasion of malware is dominated by pairwise interactions. In contrast, as the infection density increases, higher-order nonlinear effects become increasingly significant and play a decisive role in shaping the post-invasion dynamics.

4.2. Existence and stability of the endemic equilibrium

In this section, we analyze the existence and stability of endemic equilibria of system (5). An endemic equilibrium corresponds to a steady state in which the malware persists in the network at a constant, nonzero level. The possible endemic equilibria of system (5) are denoted by

$$E^* = (S^{A*}, I^{U*}, I^{A*}).$$

To determine the endemic equilibrium, we set the time derivatives in system (5) to zero, which yields the algebraic system given in equation (6). By appropriately combining these equations, the variables I^{U*} and S^{A*} can be expressed explicitly in terms of I^{A*} , as shown in equation (7):

$$\begin{cases} I^{U*} = \frac{v}{u}I^{A*} \\ S^{A*} = \frac{u}{u+v} - I^{A*}. \end{cases} \quad (7)$$

Substituting equation (7) into equation (6) reduces the equilibrium conditions to a single quadratic equation with respect to I^{A*} , given by

$$f(I^{A*}) = A_1(I^{A*})^2 + B_1I^{A*} + C_1, \quad (8)$$

where the coefficients are defined as

$$\begin{cases} A_1 = (u + v)\beta_{\Delta}\langle k_{\Delta} \rangle \\ B_1 = (u + v)\beta\langle k \rangle - u\beta_{\Delta}\langle k_{\Delta} \rangle \\ C_1 = \gamma(u + v) - u\beta\langle k \rangle. \end{cases} \quad (9)$$

Since $A_1 > 0$, equation (8) represents an upward-opening parabola. The discriminant of this quadratic equation is therefore given by

$$\begin{aligned} \Delta_1 &= B_1^2 - 4A_1C_1 \\ &= [(u + v)\beta\langle k \rangle - u\beta_{\Delta}\langle k_{\Delta} \rangle]^2 - 4[(u + v)\beta_{\Delta}\langle k_{\Delta} \rangle][\gamma(u + v) - u\beta\langle k \rangle] \\ &= \langle k \rangle^2(u + v)^2\beta^2 + 2u(u + v)\beta_{\Delta}\langle k \rangle\langle k_{\Delta} \rangle\beta + u^2\beta_{\Delta}^2\langle k_{\Delta} \rangle^2 - 4\gamma(u + v)^2\beta_{\Delta}\langle k_{\Delta} \rangle. \end{aligned} \quad (10)$$

The roots of equation (8), denoted by I_1^{A*} and I_2^{A*} , are given by

$$\begin{cases} I_1^{A*} = \frac{-B_1 - \sqrt{\Delta_1}}{2A_1} \\ I_2^{A*} = \frac{-B_1 + \sqrt{\Delta_1}}{2A_1}, \end{cases} \quad (11)$$

which correspond to the active infected densities of two distinct endemic equilibria, $E_1^* = (S_1^{A*}, I_1^{U*}, I_1^{A*})$ and $E_2^* = (S_2^{A*}, I_2^{U*}, I_2^{A*})$, respectively.

Theorem 4.2. *The number of endemic equilibria of system (5) is determined by the following conditions:*

- (i) *A unique endemic equilibrium exists if $\beta > \beta_{c1}$.*
- (ii) *Two distinct endemic equilibria exist if $0 < -\frac{B_1}{2A_1} < 1$ and $\beta_{c2} < \beta < \beta_{c1}$.*
- (iii) *A unique endemic equilibrium exists if $0 < -\frac{B_1}{2A_1} < 1$ and $\beta = \beta_{c2}$.*

where

$$\beta_{c2} = \max \left\{ 0, \frac{-u\beta_{\Delta}\langle k_{\Delta} \rangle + 2(u + v)\sqrt{\beta_{\Delta}\langle k_{\Delta} \rangle}}{(u + v)\langle k \rangle} \right\}.$$

Proof. The proof is provided in appendix A.2. □

Remark 4. The existence of multiple endemic equilibria indicates that, under identical network parameters, the system may evolve toward distinct long-term infection levels. This implies that early-stage intervention is crucial, since reducing the initial infection level can prevent the network from transitioning to a high-infection state. Such behavior highlights the importance of prompt malware detection and mitigation strategies in practical WSN deployments.

Theorem 4.3. *The endemic equilibrium $E_1^* = (S_1^{A*}, I_1^{U*}, I_1^{A*})$ is unstable.*

Proof. The detailed proof is provided in appendix A.3. □

Remark 5. The instability of the equilibrium E_1^* indicates that this state is highly sensitive to perturbations and therefore cannot be sustained as a long-term operating condition. In realistic wireless sensor networks, even small fluctuations in infection levels around this equilibrium may drive the system toward either malware extinction or persistent infection. From a control perspective, this result highlights the importance of early intervention to prevent the network from crossing a critical tipping point.

Theorem 4.4. *The equilibrium point $E_2^* = (S_2^{A*}, I_2^{U*}, I_2^{A*})$ is locally asymptotically stable if and only if the following conditions are satisfied: $P_0 > 0$, $P_1 > 0$, $P_2 > 0$, and $P_1P_2 - P_0 > 0$, where the coefficients P_0 , P_1 , and P_2 are given by*

$$\begin{cases} P_0 = 2(u + v) + \frac{1-v}{u+v} (2A_2(I_2^{A*})^2 + B_1 I_2^{A*}), \\ P_1 = (u + v)^2 + [u + (1 - v)(u + v)] (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}), \\ P_2 = u (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}). \end{cases}$$

Proof. The detailed proof is provided in appendix A.4. □

Remark 6. The stable endemic equilibrium E_2^* corresponds to a persistent operating regime in which malware persists at a nonzero level and continuously consumes network resources. Once the system converges to this equilibrium, routine recovery mechanisms may no longer be sufficient to eradicate the malware completely. This observation suggests that effective control strategies should focus on preventing the network from entering this regime, for example, by reducing higher-order infection opportunities or by enforcing stricter sleep–active scheduling policies.

Corollary 4.1. Combining the results of Theorems 4.1–4.4, system (5) exhibits the following threshold-dependent dynamics:

- (i) If $\beta < \beta_{c2}$, the disease-free equilibrium E_0 is the only stable equilibrium.
- (ii) If $\beta_{c2} < \beta < \beta_{c1}$, the system is bistable: the disease-free equilibrium E_0 and the endemic equilibrium E_2^* are both locally asymptotically stable, separated by the unstable equilibrium E_1^* .
- (iii) If $\beta > \beta_{c1}$, the disease-free equilibrium E_0 becomes unstable and the endemic equilibrium E_2^* is the only stable equilibrium.

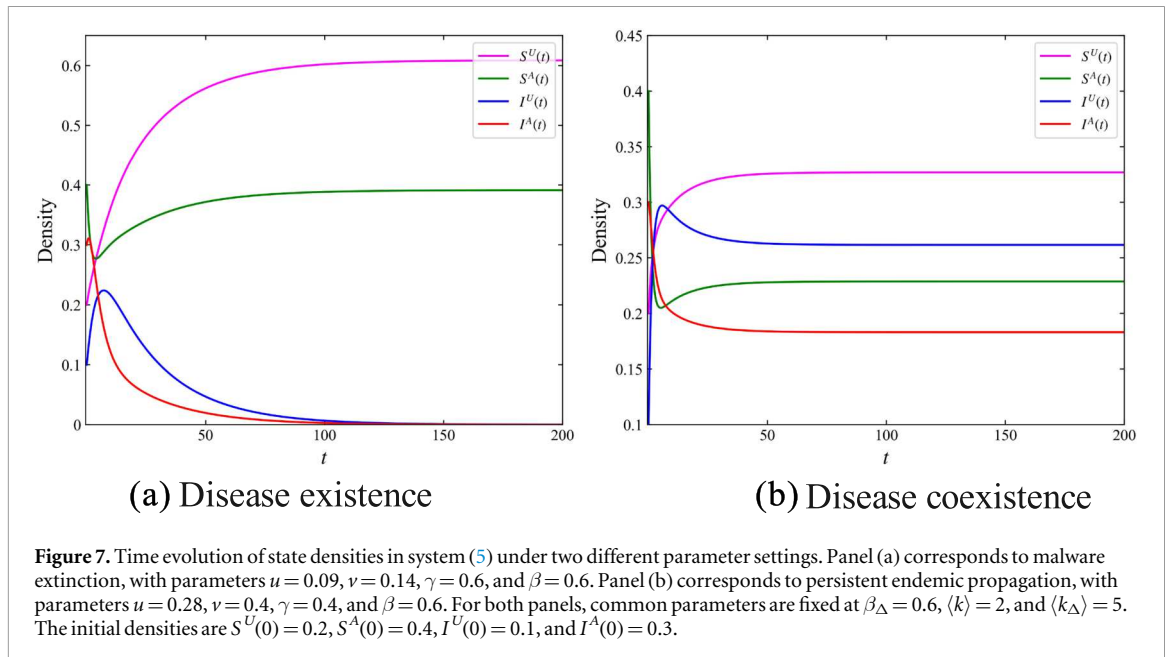
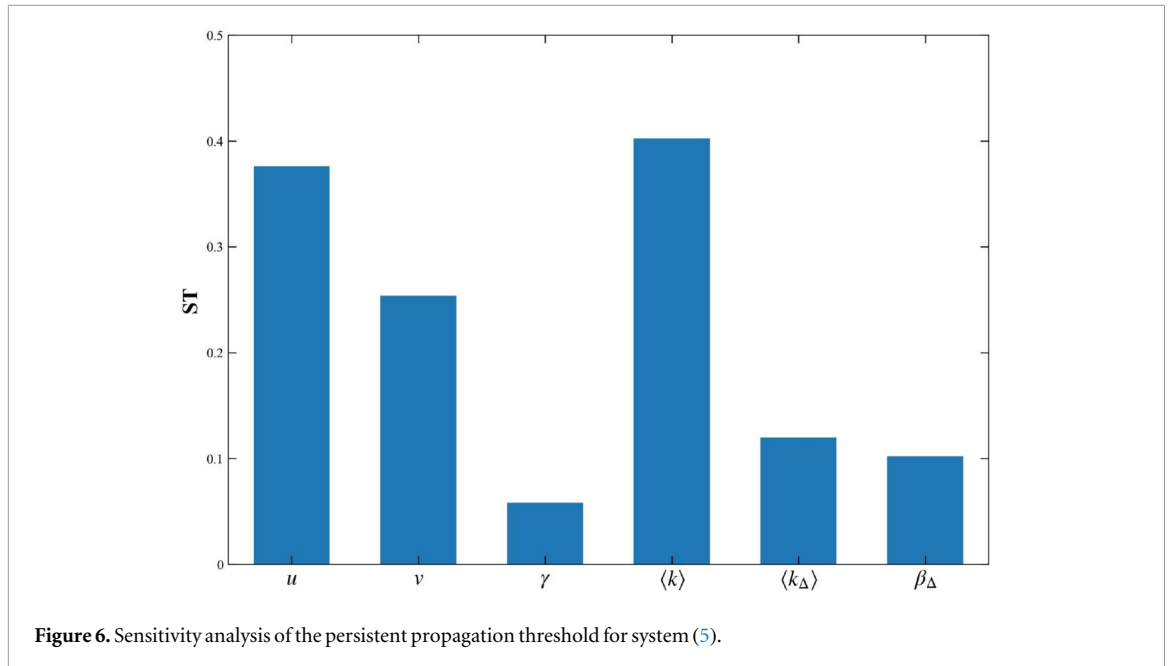
Remark 7. Corollary 4.1 shows that system (5) admits a bistable regime for $\beta_{c2} < \beta < \beta_{c1}$, where the stable equilibria E_0 and E_2^* coexist and are separated by the unstable equilibrium E_1^* . Mechanistically, bistability results from the competition between two effects: (i) sleep scheduling acts as an effective suppression mechanism by reducing the fraction of active susceptible nodes, which promotes the attraction basin of E_0 in the low-infection-density regime; (ii) higher-order infection introduces nonlinear reinforcement in the force of infection, which is negligible near the disease-free equilibrium but becomes dominant at higher infection densities, which becomes dominant at higher infection densities and can sustain the endemic state E_2^* . This competition yields initial-condition dependence and hysteresis, which is characteristic of backward bifurcation.

5. Sensitivity analysis

Sobol sensitivity analysis is a widely used global variance-based method for quantifying the relative importance of input parameters. It achieves this by decomposing the total variance of a model output into contributions from individual parameters as well as their interactions [40]. In this work, we apply this method to analyze the persistent propagation threshold of system (5), given in equation (17). This threshold plays a crucial role in characterizing the bistability and backward bifurcation behavior of the system, as it determines the conditions under which malware can persist even when the pairwise infection rate β is below the invasion threshold β_{c1} .

The total influence (ST) index is used to quantify the sensitivity, or overall influence, of a given parameter (such as the active and sleep conversion rates) on the persistent propagation threshold. To evaluate the sensitivity of β_{c2} to its constituent parameters, we set the sampling size to 1024 and computed the total influence indices of six parameters, namely u , v , γ , $\langle k \rangle$, $\langle k_\Delta \rangle$, and β_Δ , with respect to the persistent propagation threshold.

The results, shown in figure 6, indicate that the average degree $\langle k \rangle$ has the strongest impact on the persistent propagation threshold, followed by the active conversion rate u and the sleep conversion rate v . The triplewise infection rate β_Δ and the average generalized degree of order 2, $\langle k_\Delta \rangle$, exert moderate influences, whereas the recovery rate γ has the weakest effect. These findings suggest that, although recovery measures such as resetting malware-infected wireless sensors are necessary, adjusting the network topology and optimizing sleep–active scheduling strategies may provide more effective means for controlling persistent malware propagation.



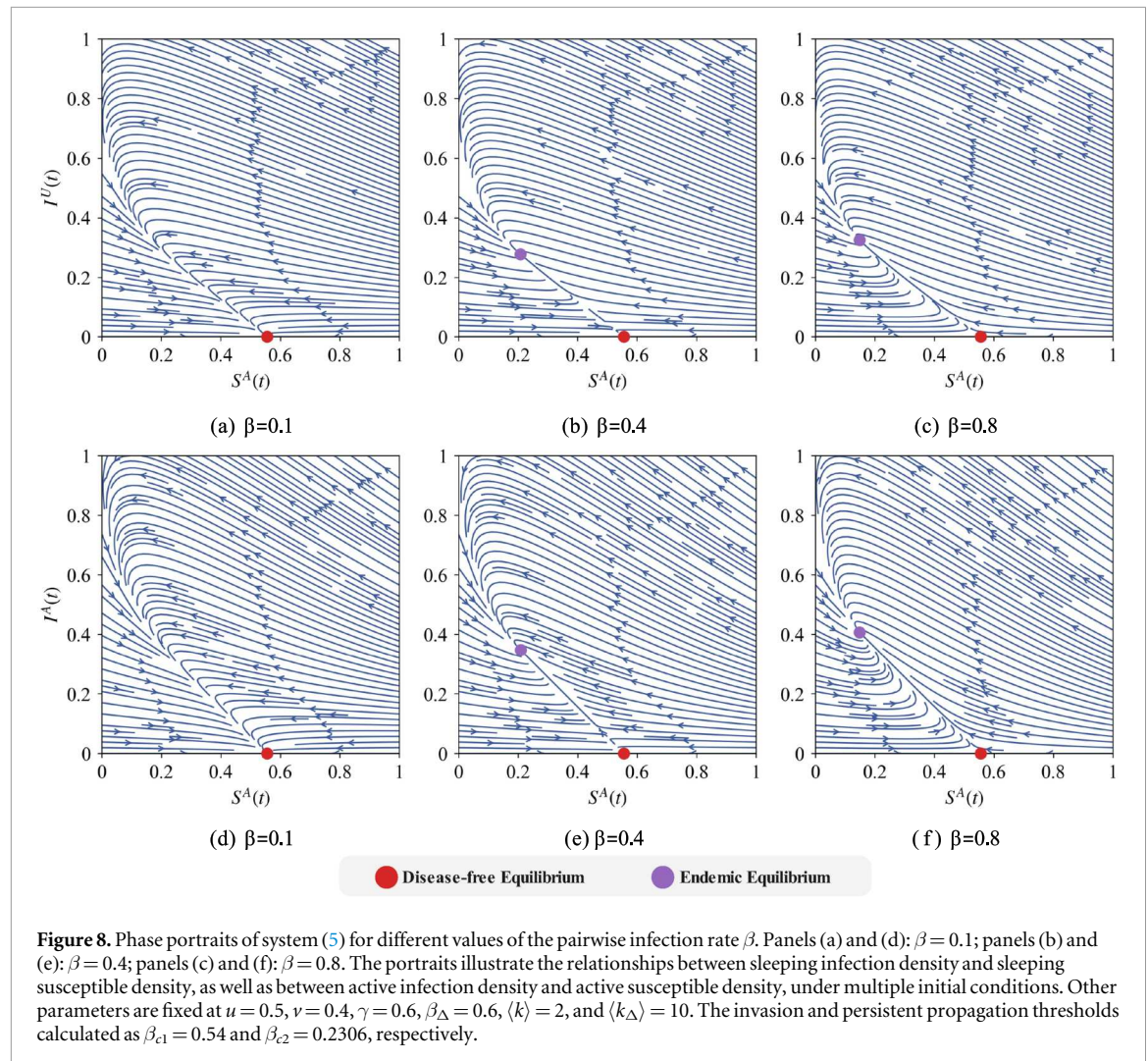
6. Numerical simulations

In this section, we conduct numerical simulations to validate the theoretical analysis. Using bifurcation diagrams and heatmaps, we further investigate how individual parameters, as well as their combined effects, influence the bifurcation structure and bistability phenomena of system (5).

6.1. Dynamic analysis and stability verification

To provide an intuitive illustration of the dynamical evolution of the system and to numerically verify the stability properties of its equilibrium points under specific parameter regimes, we perform a series of numerical simulations for system (5).

Figure 7 illustrates the temporal evolution of node state densities, which eventually converge to steady states. Panels (a) and (b) exhibit two qualitatively distinct outcomes of malware propagation in a WSN, corresponding to eventual extinction and persistent endemic spread, respectively. In panel (a), as time t increases, the density of active susceptible nodes initially decreases and subsequently increases, whereas both the sleeping and active infected densities first rise and then decline. Meanwhile, the density of sleeping susceptible nodes increases monotonically. Ultimately, the system converges to the disease-free equilibrium, at which both

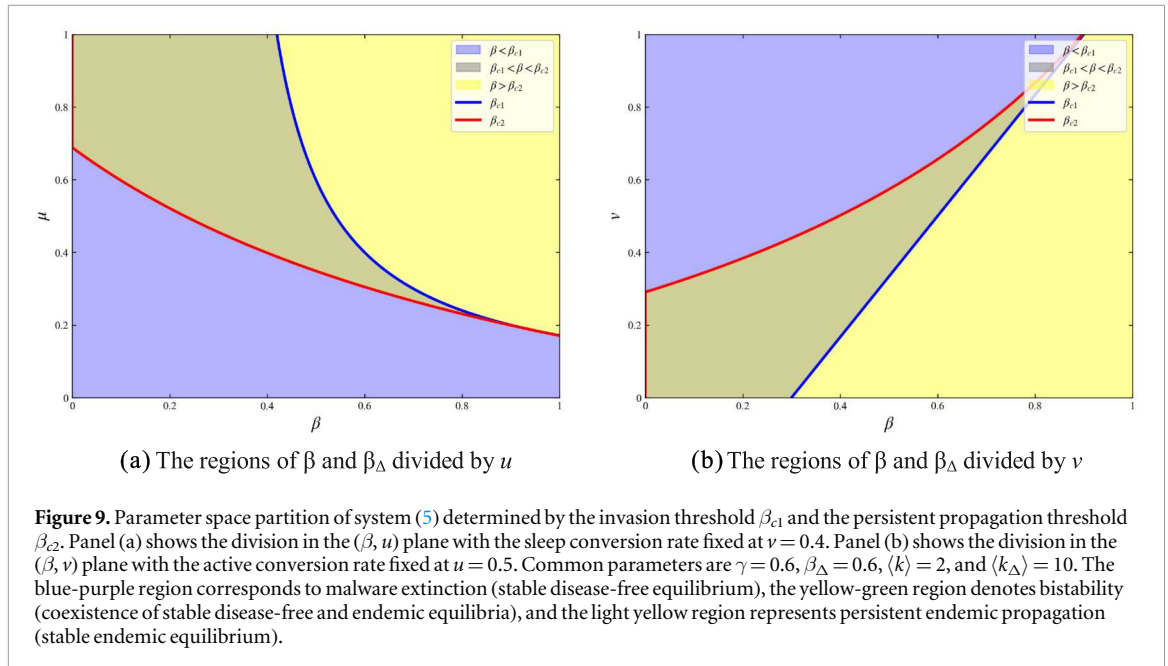


infection densities vanish, indicating the successful eradication of malware from the network. By contrast, in panel (b), the system converges to an endemic equilibrium. As time evolves, the sleeping infected density exhibits a rise followed by a decline, the active susceptible density decreases initially and then rebounds, the sleeping susceptible density increases monotonically, and the active infected density decreases monotonically. In this case, all four state densities approach distinct positive steady-state values, implying persistent malware circulation within the WSN.

Figure 8 presents the phase portraits of system (5) under different initial conditions, illustrating how the system dynamics depend on the pairwise infection rate. The phase portraits depict the relationships between the sleeping infected density and sleeping susceptible density, as well as between the active infected density and active susceptible density. When $\beta = 0.1$ (panels (a) and (d)), the system admits a single globally stable disease-free equilibrium. At an intermediate value of $\beta = 0.4$ (panels (b) and (e)), the system exhibits bistability, characterized by the coexistence of a stable disease-free equilibrium and a stable endemic equilibrium. For a high infection rate of $\beta = 0.8$ (panels (c) and (f)), only a unique stable endemic equilibrium exists. Collectively, these numerical results demonstrate that, as the pairwise infection rate increases, system (5) undergoes a bifurcation transition from a regime with only a stable disease-free equilibrium to one with only a stable endemic equilibrium, passing through an intermediate bistable region. This behavior is fully consistent with the preceding theoretical analysis on the existence and stability of the system equilibria.

The critical thresholds β_{c1} and β_{c2} represent the malware invasion threshold and the persistent propagation threshold, respectively. As shown in their analytical expressions equations (12) and (17), both thresholds depend on the active conversion rate, the sleep conversion rate, and the pairwise infection rate. Consequently, these parameters play a crucial role in determining both the malware propagation dynamics and the emergence of bistability in system (5).

Figure 9 presents bifurcation diagrams of system (5) in two different parameter planes. The threshold curves β_{c1} and β_{c2} partition each parameter plane into three regions associated with distinct dynamical behaviors. Specifically, figure 9(a) depicts the partition in the (β, u) parameter space, while figure 9(b) shows the



corresponding structure in the (β, v) parameter space. In both cases, the parameter plane is divided into three qualitatively different dynamical regimes: (i) a malware extinction region, in which the disease-free equilibrium is uniquely stable; (ii) a bistable region, where a stable disease-free equilibrium coexists with a stable endemic equilibrium, implying that the malware may either be eradicated or persist endemically depending on initial conditions; and (iii) an endemic region, where a unique stable endemic equilibrium exists, corresponding to persistent malware propagation in the WSN.

Notably, these three dynamical regimes are fully consistent with the phase-portrait behaviors illustrated in figure 8. Specifically, the disease-free, bistable, and endemic regimes correspond to the dynamics shown in figures 8(a) and (d), (b) and (e), and (c) and (f), respectively.

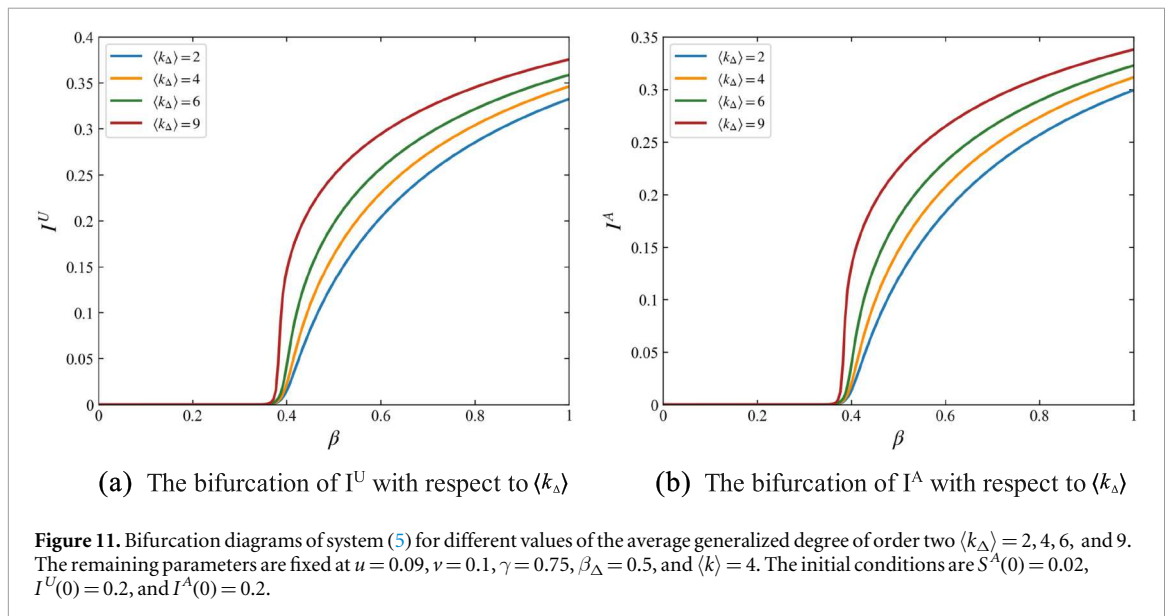
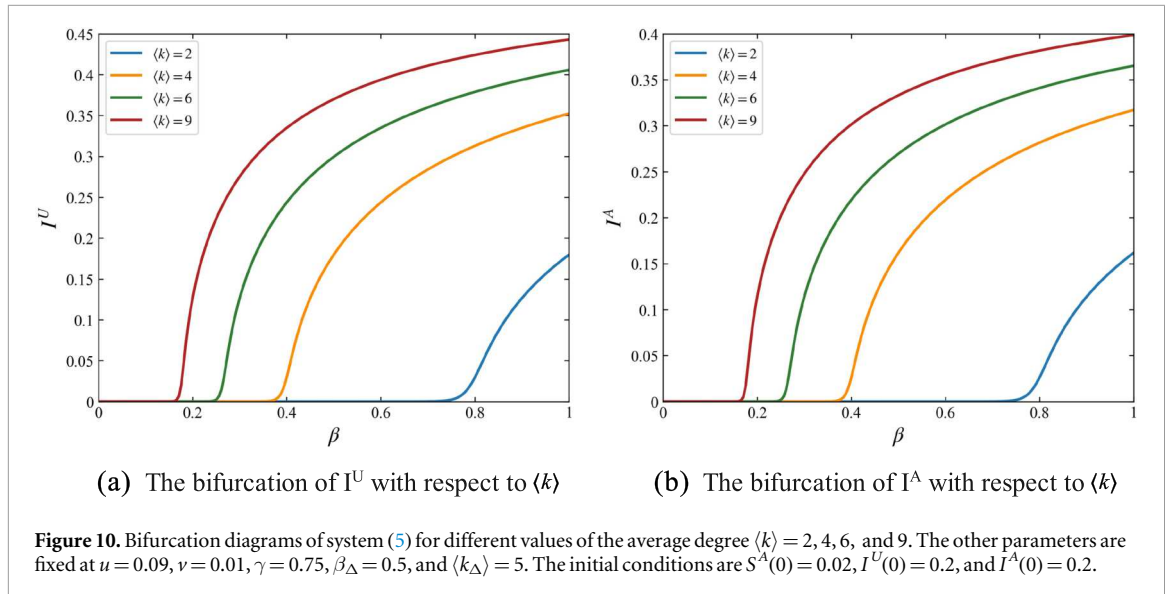
6.2. Bifurcation analysis and bistability

This subsection examines the effect of varying a single control parameter on the bifurcation structure of system (5), with a particular focus on the emergence of bistability associated with backward bifurcation.

In traditional complex network models, where infection spreads exclusively through pairwise interactions, the basic reproduction number R_0 serves as a critical threshold equal to unity. When $R_0 < 1$, the system admits a unique, locally asymptotically stable disease-free equilibrium (E_0), whereas for $R_0 > 1$, a unique and stable endemic equilibrium (E^*) emerges. The qualitative transition from a stable disease-free state to a stable endemic state as R_0 crosses unity is referred to as a forward bifurcation. By contrast, networks incorporating higher-order interactions can exhibit substantially richer dynamical behaviors. Although a unique endemic equilibrium E^* typically still exists when $R_0 > 1$, a fundamental qualitative difference arises in the regime $R_0 < 1$. Under certain conditions, the system may exhibit bistability, in which a stable disease-free equilibrium coexists with a stable endemic equilibrium. The appearance of such a bistable region allows the persistence of an endemic state even when the basic reproduction number is below unity, and this phenomenon is a defining hallmark of backward bifurcation.

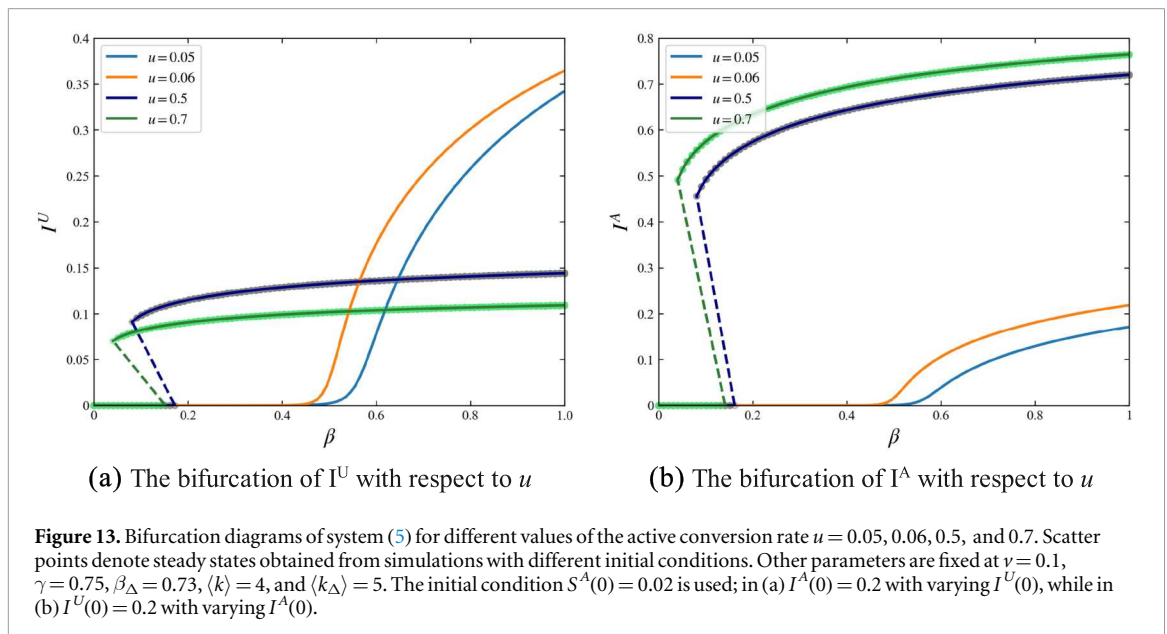
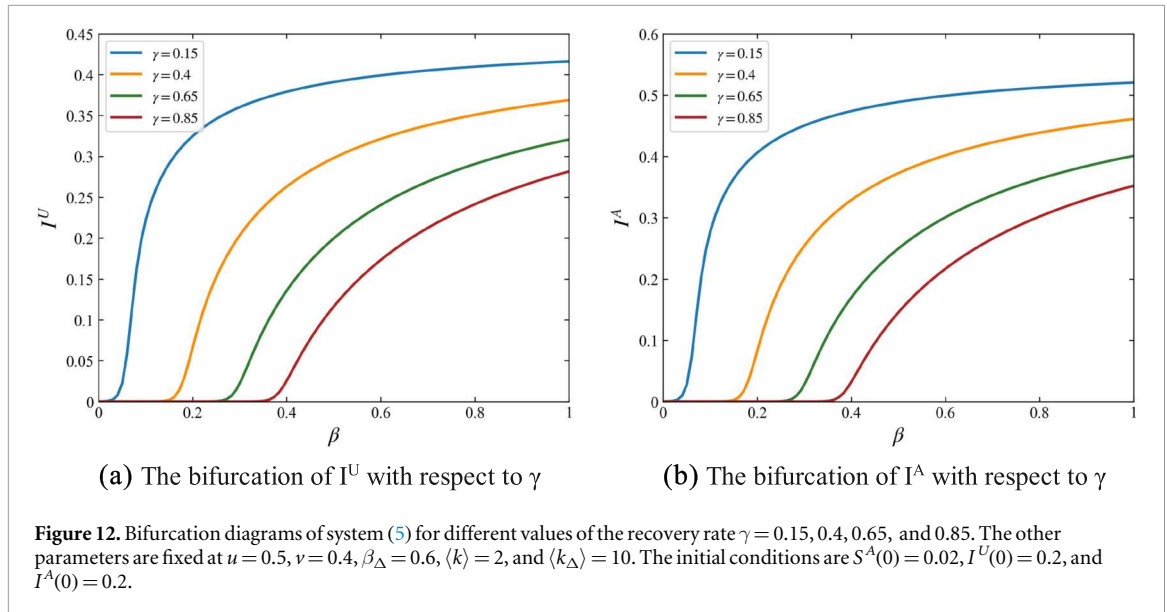
The impact of various parameters on the forward and backward bifurcation dynamics of system (5) is examined next. To this end, we analyze bifurcation diagrams that depict the steady-state values of the sleeping and active infection densities. As shown in figure 10, increasing the average node degree leads to higher steady-state levels of both sleeping and active infections, while simultaneously lowering the critical malware invasion threshold. This effect facilitates an earlier onset of large-scale malware invasion. Moreover, a comparison between figures 10(a) and (b) indicates that, under identical parameter settings, the steady-state sleeping infection density consistently exceeds the corresponding active infection density.

As shown in figure 11, variations in the average generalized degree of order two $\langle k_{\Delta} \rangle$ do not alter the bifurcation type of system (5). An increase in $\langle k_{\Delta} \rangle$ raises the endemic steady-state levels of both sleeping and active infection densities, while leaving the invasion threshold unchanged. This result indicates that higher-order connectivity influences the magnitude of post-invasion persistence but does not affect the qualitative bifurcation structure.



The influence of the recovery rate γ on the bifurcation behavior of system (5) is examined in figure 12. As γ increases, both the sleeping and active infection densities decrease, while the malware invasion threshold shifts toward higher values. This indicates that enhanced recovery mechanisms suppress endemic persistence and delay the onset of large-scale malware propagation. These results suggest that timely antivirus updates or an increased frequency of hardware resets can effectively reduce the density of malware-infected sensors in WSNs and postpone malware invasion. In addition, under identical parameter settings, the steady-state sleeping infection density is consistently higher than the active infection density.

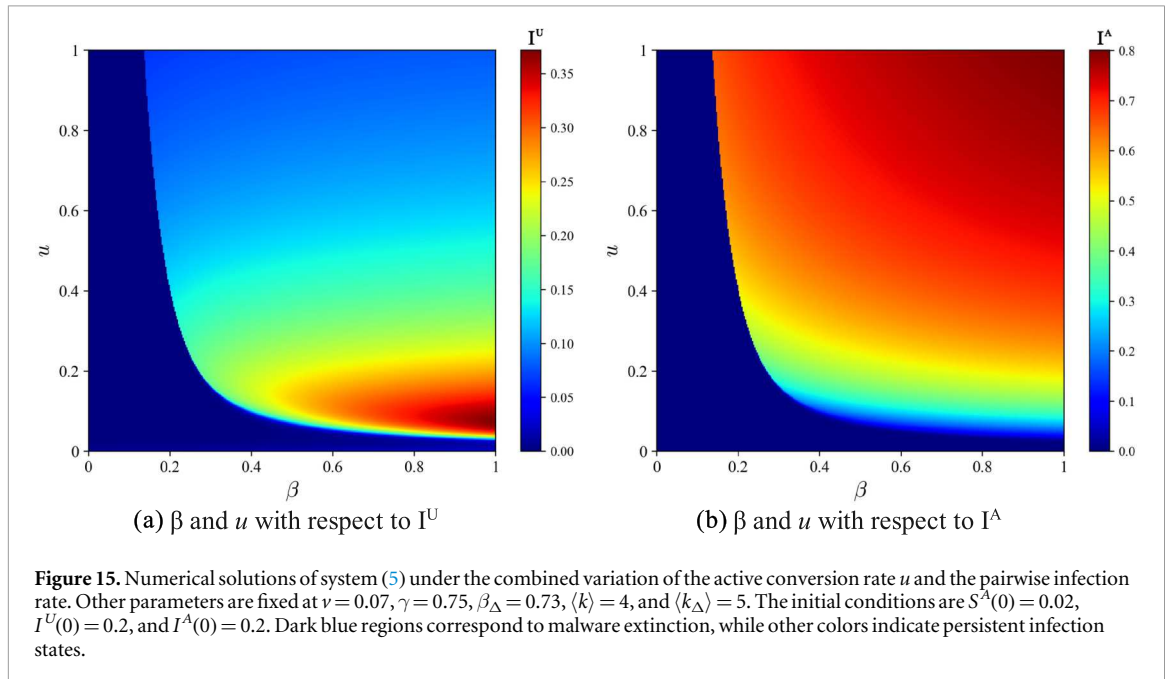
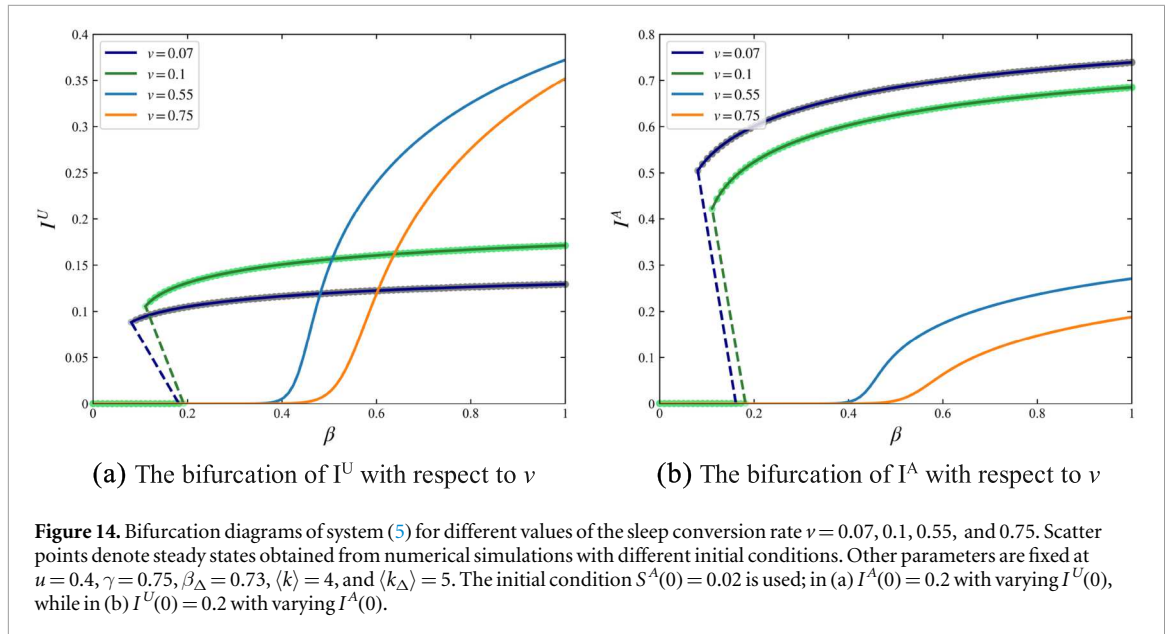
Figure 13 examines the effect of the active conversion rate on the bifurcation behavior of the steady-state densities of the sleeping-infected and active-infected states in system (5). The solid curves represent the bifurcation diagrams corresponding to different values of u . The light blue and orange curves correspond to relatively small active conversion rates ($u = 0.05$ and $u = 0.06$, respectively), for which the system exhibits a forward bifurcation. By contrast, the dark blue and green curves correspond to larger active conversion rates ($u = 0.5$ and $u = 0.7$, respectively), under which the system undergoes a backward bifurcation. In addition, under identical parameter settings, the steady-state sleeping infection density is consistently lower than the active infection density. The dashed curves connect the invasion threshold and the persistent propagation threshold, thereby delineating the bistable region. As the active conversion rate increases, the system transitions from forward to backward bifurcation, accompanied by a decrease in both the malware invasion threshold and the persistent propagation threshold. The distribution of scatter points in figure 13 further confirms that the system may converge to different steady states depending on the initial conditions. In particular, within the bistable regime,



malware propagation in WSNs may either die out or persist endemically, with the eventual outcome determined by the initial infection densities. This observation indicates that the long-term behavior of the system depends not only on the active conversion rate but also critically on the initial values of $I^U(0)$ and $I^A(0)$.

Figure 14 investigates the influence of the sleep conversion rate on the bifurcation behavior of the steady-state densities of the sleeping-infected and active-infected states in system (5). The solid curves depict the bifurcation diagrams corresponding to different values of the sleep conversion rate. For relatively large sleep conversion rates, the system exhibits a forward bifurcation. In contrast, when the sleep conversion rate is sufficiently small, the system undergoes a backward bifurcation. As the sleep conversion rate decreases, both the malware invasion threshold β_{c1} and the persistent propagation threshold shift toward smaller values, indicating that a reduced tendency for nodes to enter the sleep state facilitates sustained malware propagation. Moreover, under identical parameter settings, the steady-state sleeping infection density is consistently lower than the active infection density. The dashed curves connect the invasion threshold and the persistent propagation threshold, thereby delineating the bistable region. The distribution of scatter points further confirms that, within this bistable regime, distinct initial conditions may lead the system to different steady states. This observation demonstrates that the long-term dynamics of the system depend not only on the sleep conversion rate, but also critically on the initial densities of sleeping and active infected nodes.

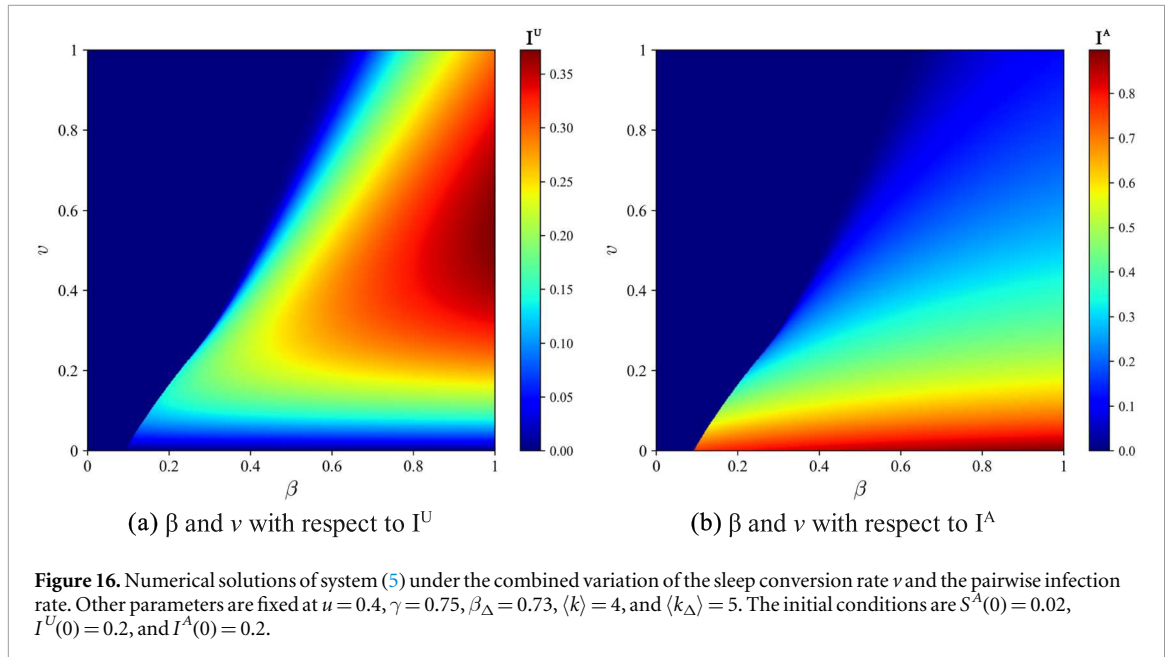
A reduction in the active conversion rate together with an increase in the sleep conversion rate can raise both the malware invasion threshold and the persistent propagation threshold. This indicates that, while



maintaining normal WSN operation, decreasing the active conversion rate or increasing the sleep conversion rate can not only effectively suppress malware propagation but also reduce energy consumption, thereby extending the operational lifetime of the network.

6.3. Synergistic effect of two parameters on steady state

To investigate the synergistic effects of parameter pairs on system (5), figures 15 and 16 present heat maps of the corresponding steady-state outcomes. In these plots, regions associated with the lowest values on the color scale, shown in dark blue, indicate malware extinction, whereas regions with other colors correspond to persistent malware infection. The boundary separating the dark blue region from the remaining colored regions represents the malware invasion threshold. Figure 15 illustrates the combined effects of the active conversion rate and the pairwise infection rate on the steady-state densities of the sleeping-infected and active-infected states in system (5). It can be observed that the malware invasion threshold decreases as the active conversion rate increases, indicating that more frequent transitions to the active state facilitate malware invasion. Within the endemic region, for a fixed pairwise infection rate, the sleeping infection density exhibits a nonmonotonic dependence on the active conversion rate, first increasing and then decreasing as u increases. In contrast, the active infection density increases monotonically with the active conversion rate. These results



indicate that although reducing the active conversion rate is an effective strategy for suppressing active infections and enhancing network security, its impact on the sleeping infection density must also be taken into account. Moreover, the bistable region gradually shrinks and eventually disappears as the active conversion rate decreases.

Figure 16 illustrates the combined effects of the sleep conversion rate and the pairwise infection rate on the steady-state sleeping and active infection densities in system (5). It can be observed that increasing the sleep conversion rate raises the malware invasion threshold, indicating that more frequent transitions to the sleeping state hinder malware invasion. Within the endemic region, for a fixed pairwise infection rate, the sleeping infection density exhibits a nonmonotonic dependence on the sleep conversion rate, first increasing and then decreasing as ν increases. In contrast, the active infection density decreases monotonically with the sleep conversion rate. Notably, in panel (a), when the sleep conversion rate is very small, the sleeping infection density approaches zero, since only a negligible fraction of active nodes transition into the sleeping state.

7. Conclusion

In this paper, we propose a malware propagation model for WSNs that incorporates higher-order interactions and node sleep-scheduling. The model captures two essential features of malware spreading in WSNs: the sleep-active scheduling mechanism of sensor nodes and the coexistence of pairwise transmission and higher-order group-based infections. Through mean-field approximation, our theoretical analysis reveals that the system can exhibit a backward bifurcation, leading to a bistable regime in which a stable disease-free equilibrium coexists with a stable endemic equilibrium. This implies that large-scale malware outbreaks may still occur even when the pairwise infection rate remains below the invasion threshold. Numerical simulations further validate the existence of this bistability and demonstrate its strong dependence on key system parameters. In particular, a lower active conversion rate or a higher sleep conversion rate can induce backward bifurcation, giving rise to a bistable region in which the final steady state of the system depends sensitively on the initial infection density.

This study has two main limitations. First, the proposed model assumes a static network topology, whereas real-world wireless sensor networks may exhibit time-varying structures. Second, owing to the relatively low probability of higher-order infection events, the analysis is restricted to 2-simplicial complexes and does not consider infections arising from interactions on higher-order simplices. These limitations point to several meaningful directions for future research, including extending the model to dynamic networks and investigating the role of higher-order interactions on simplicial structures of higher dimensions. Overall, this study elucidates how node sleep-active scheduling and higher-order interactions jointly shape the dynamical evolution of malware propagation in WSNs, and it provides valuable theoretical insights for related research areas.

Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

Appendix

A.1. Proof of theorem 4.1

To calculate the basic reproduction number using the next-generation matrix method [41], we only need to consider the equations associated with the infected compartments, namely $\frac{dI^U(t)}{dt}$ and $\frac{dI^A(t)}{dt}$, in system (5). Accordingly, we define the new infection matrix F and the state transition matrix V as follows:

$$F = \begin{pmatrix} 0 & vM \\ 0 & (1 - v)M \end{pmatrix},$$

and

$$V = \begin{pmatrix} u & v(\gamma - 1) \\ -u & v + \gamma - v\gamma \end{pmatrix},$$

where $M = \beta \langle k \rangle S^A(t) + 2\beta_\Delta \langle k_\Delta \rangle S^A(t) I^A(t)$.

Based on the previously defined matrices F and V , the inverse matrix V^{-1} and the resulting next-generation matrix FV^{-1} are computed as follows:

$$\begin{cases} V^{-1} = \frac{1}{u\gamma} \begin{pmatrix} v + \gamma - v\gamma & v(1 - \gamma) \\ u & u \end{pmatrix} \\ FV^{-1} = \frac{M}{\gamma} \begin{pmatrix} v & v \\ 1 - v & 1 - v \end{pmatrix}. \end{cases}$$

The basic reproduction number R_0 is defined as the spectral radius, that is, the largest eigenvalue, of the next-generation matrix FV^{-1} derived above. Accordingly, R_0 is obtained by solving the characteristic equation of FV^{-1} , which is given by

$$\begin{aligned} |\xi I - FV^{-1}| &= \begin{vmatrix} \xi - \frac{vM}{\gamma} & -\frac{vM}{\gamma} \\ -\frac{(1-v)M}{\gamma} & \xi - \frac{(1-v)M}{\gamma} \end{vmatrix} \\ &= \xi \left(\xi - \frac{M}{\gamma} \right). \end{aligned}$$

Therefore, the basic reproduction number R_0 is obtained as

$$R_0 = \frac{M}{\gamma} \Big|_{E_0} = \frac{u\beta \langle k \rangle}{\gamma(u + v)}.$$

The malware invasion threshold β_{c1} is obtained by setting $R_0 = 1$, which yields

$$\beta_{c1} = \frac{\gamma(u + v)}{u \langle k \rangle}. \quad (12)$$

The malware invasion threshold β_{c1} determines the stability of the disease-free equilibrium E_0 : E_0 is locally asymptotically stable for $\beta < \beta_{c1}$ (corresponding to $R_0 < 1$) and becomes unstable for $\beta > \beta_{c1}$ (corresponding to $R_0 > 1$).

A.2. Proof of theorem 4.2

The existence of endemic equilibria is determined by the number of positive real roots I^{A*} of the quadratic equation (8) lying in the interval $[0, 1]$. We analyze this dependence with respect to the pairwise infection rate β .

From equations (12), (8), and (9), the expressions of $f(0)$ and $f(1)$ are given in equation (13) as follows:

$$\begin{cases} f(0) = C_1 = \gamma(u + v) - u\beta \langle k \rangle = u \langle k \rangle (\beta_{c1} - \beta) \\ f(1) = A_1 + B_1 + C_1 = v\beta \langle k \rangle + v\beta_\Delta \langle k_\Delta \rangle + \gamma(u + v). \end{cases} \quad (13)$$

When $\beta > \beta_{c1}$, it follows from equation (13) that $f(0) < 0$ and $f(1) > 0$. Since the function $f(I^{A*})$, defined in equation (8), is a convex quadratic function, these conditions guarantee the existence of a unique root in the interval $[0, 1]$. Therefore, for $\beta > \beta_{c1}$, the system admits a unique endemic equilibrium.

To analyze the system under other conditions, we examine the discriminant of the quadratic equation (8), given by

$$\begin{aligned}\Delta_1(\beta) &= B_1^2 - 4A_1C_1 \\ &= A_2\beta^2 + B_2\beta + C_2.\end{aligned}\quad (14)$$

The coefficients in equation (14) are defined as

$$\begin{cases} A_2 = \langle k \rangle^2 (u + v)^2 \\ B_2 = 2u(u + v)\beta_\Delta \langle k \rangle \langle k_\Delta \rangle \\ C_2 = u^2\beta_\Delta^2 \langle k_\Delta \rangle^2 - 4\gamma(u + v)^2\beta_\Delta \langle k_\Delta \rangle. \end{cases}\quad (15)$$

Applying the quadratic formula to equation (14), the roots of the equation $\Delta_1(\beta) = 0$ are given by

$$\begin{cases} \beta_{c2} = \frac{-B_2 + \sqrt{\Delta_2}}{2A_2} \\ \beta_{c3} = \frac{-B_2 - \sqrt{\Delta_2}}{2A_2}, \end{cases}\quad (16)$$

where $\Delta_2 = B_2^2 - 4A_2C_2$.

Since $A_2 > 0$ and $B_2 > 0$, it follows that the root β_{c3} is negative and not physically meaningful. Accordingly, we only retain the critical value β_{c2} , which is given by

$$\begin{aligned}\beta_{c2} &= \max \left\{ 0, \frac{-B_2 + \sqrt{\Delta_2}}{2A_2} \right\} \\ &= \max \left\{ 0, \frac{-u\beta_\Delta \langle k_\Delta \rangle + 2(u + v)\sqrt{\beta_\Delta \langle k_\Delta \rangle}}{(u + v)\langle k \rangle} \right\}.\end{aligned}\quad (17)$$

For the interval $\beta_{c2} < \beta < \beta_{c1}$, it follows from equation (13) that $f(0) > 0$ and $f(1) > 0$. Moreover, since $A_2 > 0$, the discriminant $\Delta_1(\beta)$ is an upward-opening parabola with β_{c2} as its positive root, which implies that $\Delta_1 > 0$ in this interval. Because equation (8) also represents an upward-opening quadratic function, the condition $0 < -\frac{B_1}{2A_1} < 1$ ensures that the axis of symmetry lies within the interval $(0, 1)$. Finally, when $\beta = \beta_{c2}$, the discriminant satisfies $\Delta_1 = 0$. From equation (13), we obtain $f(0) > 0$ and $f(1) > 0$. Since equation (8) represents an upward-opening quadratic function, and the condition $0 < -\frac{B_1}{2A_1} < 1$ ensures that the axis of symmetry lies within the interval $(0, 1)$, the equation admits a unique positive root in $(0, 1)$. Consequently, there exists a unique endemic equilibrium in this case.

A.3. Proof of theorem 4.3

The Jacobian matrix of system (5) evaluated at an endemic equilibrium E^* is given by equation (18):

$$J(E^*) = \begin{pmatrix} J_{11} & J_{12} & J_{13} \\ J_{21} & J_{22} & J_{23} \\ J_{31} & J_{32} & J_{33} \end{pmatrix}.\quad (18)$$

The corresponding entries of the Jacobian matrix are explicitly given by equation (19):

$$\begin{cases} J_{11} = -(u + v) - (1 - v)\beta \langle k \rangle I^{A*} - (1 - v)\beta_\Delta \langle k_\Delta \rangle (I^{A*})^2, \\ J_{12} = -u, \\ J_{13} = \gamma - u - v\gamma - (1 - v)\beta \langle k \rangle S^{A*} - 2(1 - v)\beta_\Delta \langle k_\Delta \rangle S^{A*} I^{A*}, \\ J_{21} = v\beta \langle k \rangle I^{A*} + v\beta_\Delta \langle k_\Delta \rangle (I^{A*})^2, \\ J_{22} = -u, \\ J_{23} = v(1 - \gamma) + v\beta \langle k \rangle S^{A*} + 2v\beta_\Delta \langle k_\Delta \rangle S^{A*} I^{A*}, \\ J_{31} = (1 - v)\beta \langle k \rangle I^{A*} + (1 - v)\beta_\Delta \langle k_\Delta \rangle (I^{A*})^2, \\ J_{32} = u, \\ J_{33} = v\gamma - v - \gamma + (1 - v)\beta \langle k \rangle S^{A*} + 2\beta_\Delta \langle k_\Delta \rangle S^{A*} I^{A*}. \end{cases}\quad (19)$$

Combining equations (7) and (9), and using the fact that I_1^{A*} is a root of equation (8), the determinant of the Jacobian matrix at E_1^* can be simplified as follows:

$$\begin{aligned}
|J(E_1^*)| &= u \left[-\gamma(u + v) + u\beta\langle k \rangle - 2(u + v)\beta\langle k \rangle I_1^{A*} + 2u\beta_\Delta\langle k_\Delta \rangle I_1^{A*} \right. \\
&\quad \left. - 3(u + v)\beta_\Delta\langle k_\Delta \rangle (I_1^{A*})^2 \right] \\
&= (-u) \left[A_1 (I_1^{A*})^2 + B_1 I_1^{A*} + C_1 + 2A_1 (I_1^{A*})^2 + B_1 I_1^{A*} \right] \\
&= (-u) I_1^{A*} (2A_1 I_1^{A*} + B_1) \\
&= (-u) I_1^{A*} \left(2A_1 \frac{-B_1 - \sqrt{\Delta_1}}{2A_1} + B_1 \right) \\
&= u I_1^{A*} \sqrt{\Delta_1}.
\end{aligned} \tag{20}$$

Given that $I_1^{A*} \in [0, 1]$ and $\sqrt{\Delta_1} > 0$, it follows that the determinant of the Jacobian matrix at E_1^* is positive, that is, $|J(E_1^*)| > 0$. Let θ_1, θ_2 , and θ_3 denote the eigenvalues of the 3×3 Jacobian matrix $J(E_1^*)$. Since the product of the eigenvalues equals the determinant, we have

$$\theta_1 \theta_2 \theta_3 = |J(E_1^*)| > 0. \tag{21}$$

This implies that at least one eigenvalue of $J(E_1^*)$ is positive. Therefore, according to the Routh–Hurwitz stability criterion [42], the equilibrium $E_1^* = (S_1^{A*}, I_1^{U*}, I_1^{A*})$ is unstable.

A.4. Proof of theorem 4.4

The stability of the equilibrium E_2^* is determined by the roots of its characteristic equation, given by

$$|\zeta I - J(E_2^*)| = 0. \tag{22}$$

The characteristic equation of system (5) is derived from the Jacobian matrix given in equation (18) and (19). By combining equations (7) and (9), the characteristic equation can be written in the following cubic form:

$$\begin{aligned}
&\zeta^3 + \left[2(u + v) + \frac{1-v}{u+v} (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}) \right] \zeta^2 \\
&+ \left[(u + v)^2 + (u + (1-v)(u + v)) (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}) \right] \zeta \\
&+ u (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}) = 0.
\end{aligned} \tag{23}$$

Comparing equation (23) with the standard cubic form $\zeta^3 + P_2\zeta^2 + P_1\zeta + P_0 = 0$, the coefficients are identified as

$$\begin{cases} P_2 = 2(u + v) + \frac{1-v}{u+v} (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}), \\ P_1 = (u + v)^2 + [u + (1-v)(u + v)] (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}), \\ P_0 = u (2A_1(I_2^{A*})^2 + B_1 I_2^{A*}). \end{cases} \tag{24}$$

According to the Routh–Hurwitz stability criterion for a third-order polynomial, all roots have negative real parts if and only if $P_2 > 0, P_1 > 0, P_0 > 0$, and $P_1 P_2 - P_0 > 0$. Therefore, the equilibrium $E_2^* = (S_2^{A*}, I_2^{U*}, I_2^{A*})$ is locally asymptotically stable under these conditions.

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